

ABHISHEK KUMAR

Machine Learning & AI Engineer

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PROFESSIONAL SUMMARY

Machine Learning & AI Engineer with 2.5+ years of experience designing, optimizing, and deploying production-grade AI systems across Natural Language Processing (NLP), Computer Vision, Audio/Speech processing, and Data Engineering. Proven track record of building real-time speaker identification systems, advanced OCR-driven RAG pipelines, speech recognition (STT), and text-to-speech (TTS) models using transformer-based neural network architectures. Achieved measurable business impact by reducing manual workflows by up to 65%, improving model accuracy, and architecting scalable data pipelines, REST API microservices, and containerization strategies for enterprise deployment. Led end-to-end model training and model evaluation cycles across clustering, classification, and regression tasks. Delivered production-grade ML solutions that analyzed large-scale datasets and extracted actionable intelligence. Core technologies: LangChain, LangGraph, LlamaIndex, Flask, FastAPI, PyTorch, Whisper, YOLO, Docker, vector databases, and AWS.

PROFESSIONAL EXPERIENCE

Softel Solutions Pvt. Ltd | Python Developer (AI/ML)

Mar 2026 – Present

Noida, India

- ▶ **Real-time Speaker Identification Module:** Architected and deployed a real-time speaker training and identification system using **PyTorch**, **ECAPA-TDNN (Wav2Vec2)**, and **ChromaDB (vector database)**. Created a neural network pipeline that processed voice embeddings on-the-fly, achieving 94% accuracy across 250+ individuals.
- ▶ **Speech-to-Text (STT) & Text-to-Speech (TTS):** Built optimized audio microservices using **Whisper** and **Bark/Coqui TTS**, containerizing services via **FastAPI**. Containerized all services via Docker for scalable production deployment.
- ▶ **OCR-Driven Advanced RAG Pipeline:** Created a document intelligence system integrating **PaddleOCR**, **Tesseract**, and **LlamaIndex**. Extracted and structured text metadata from scanned PDFs, enabling conversational RAG querying and reducing search time by 35%.
- ▶ **Scalable Backend & Data Pipeline:** Led and orchestrated complex multimodal data pipelines using **LangChain** and MongoDB. Improved end-to-end pipeline throughput by 40%.

Azine Web Technologies Pvt. Ltd | Machine Learning Engineer

Feb 2025 – Feb 2026

Ahmedabad, India

- ▶ **AML Risk Assessment:** Designed and deployed an automated risk-profiling system using DistilBERT, SVM, and NLP-based entity matching, processing 30,000+ watchlist entities.
- ▶ **Compliance Automation:** Integrated 200+ OpenSanctions datasets and 150+ live news feeds, achieving a 28% increase in high-risk detection. Reduced manual compliance effort by 65%.
- ▶ **News Intelligence Classification:** Analyzed and classified 10K+ news articles weekly using multiclass crime classification and sentiment analysis, improving accuracy by 21%.
- ▶ **Identity Verification:** Implemented and deployed secure real-time identity verification APIs (liveness detection, signature recognition) using **YOLOv11** and **FastAPI**.
- ▶ **Barcode Extraction:** Engineered a document intelligence pipeline using **YOLOv11**, OpenCV, and OCR. Achieved 95%+ field extraction accuracy under 30% image degradation, reducing manual review.
- ▶ **ETL Pipelines:** Delivered robust ETL pipelines scraping 150+ government portals using Selenium, BeautifulSoup, and PDFPlumber on scheduled 3-day cron cycles.

Phoenix Labs Global | AI & ML Intern

Jul 2024 – Jan 2025

Remote

- ▶ **RAG Compliance Chatbot:** Built a RAG query system using **LangChain**, **LlamaIndex**, and MySQL for conversational audits over fintech records, reducing compliance review cycles by 40%.

- ▶ **API Orchestration Gateway:** Built a Django-based API gateway managing 13+ FastAPI ML microservices with RBAC, logging 10K+ cycles/month.
- ▶ **Transaction Monitoring & Fraud Detection:** Designed a rule-based anomaly detection engine, improved fraud flagging by reducing false positives by 37% and increased alert accuracy by 22%. Analyzed transaction patterns across diverse risk profiles.

Atharvo India Pvt. Ltd | *Machine Learning Intern*

Jan 2024 – Jun 2024

Remote

- ▶ **Churn Prediction:** Developed and trained ML models (XGBoost, Random Forest, Scikit-learn) to predict customer churn, achieving 91% accuracy and saving 15% retention costs.
- ▶ **NLP Classification:** Engineered a text classification pipeline using TF-IDF and Logistic Regression, improving ticket routing efficiency by 30%.
- ▶ **Feature Engineering:** Engineered data preprocessing pipelines using Pandas and NumPy to handle missing data, outlier detection, and feature scaling across 100K+ records.

TECHNICAL SKILLS

Languages	Python (Advanced) • Java • JavaScript • SQL
Audio & Speech	Wav2Vec2 • ECAPA-TDNN • Whisper (STT) • Bark (TTS) • Coqui TTS • Audio Diarization • Speaker Recognition
Generative AI & LLMs	LangChain • LangGraph • LlamaIndex • RAG (Retrieval-Augmented Generation) • Multi-Agent Systems • Vector Embeddings • Transformer Architecture
ML & NLP	PyTorch • TensorFlow • Scikit-learn • BERT • DistilBERT • Neural Network • Model Training • Model Evaluation • Clustering • Classification • Regression • Feature Engineering • Data Pipeline • Sentiment Analysis • Natural Language Processing • TF-IDF • NLTK
Computer Vision & OCR	YOLO • OpenCV • PaddleOCR • Tesseract • Image Processing • Object Detection • Object Classification
Backend & DevOps	FastAPI • Django • Flask • REST API • Containerization (Docker) • AWS (EC2, S3) • ChromaDB (Vector Database) • MongoDB • MySQL • Git

EDUCATION

Bachelor of Technology (B.Tech) – Electronics & Communication Engineering

Sep 2020 – Jun 2024

CGPA: 8.17 / 10

Oriental Institute of Science and Technology, Bhopal, India

CERTIFICATIONS

- ▶ **AWS & Cloud:** Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate (Mar 2025)
- ▶ **NLP & Deep Learning:** NLP Mastery in Python – Udemy (Jul 2025), Deep Learning for Developers – Infosys (Aug 2024)
- ▶ **Machine Learning:** ML Internship – Infosys Springboard (Dec 2024), Data Science & ML BootCamp – Udemy (Jul 2025)
- ▶ **Core Development:** Java & Python Full Stack – (Dec 2024), SQL for Data Science – UC Davis/Coursera (May 2023)

EXTRA-CURRICULAR ACTIVITIES

- ▶ **Event Management Intern – Aashman Foundation:** Managed logistics for 6+ community events, increasing attendee engagement by 25%.
- ▶ **Vice-Captain – College Cricket Team:** Guided a 16-member team through 10+ tournaments, achieving 2 state-level podium finishes.